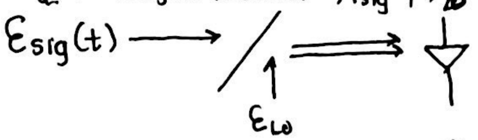
EE558 HW4 Binh Phan

1. a. Left is NRZ and right is RZ because RZ needs more peak power and BW

b. NRZ is more tolerant to dispersion (less peak power)

c. pulse time $\Rightarrow \tau \rightarrow$ distance between each peak $= \frac{1}{\tau}$

2. a. Heterodyne receiver: $\lambda_{sig} \neq \lambda_{LO}$, we mix the incoming data signal with a powerful local oscillator.



Since $I \propto |\vec{\mathcal{E}}|^2$, $I(t) = |\vec{\mathcal{E}}_{sig}(t) + \mathcal{E}_{LO}|^2 = |\mathcal{E}_0(t)(\cos\omega_s(t) + \phi_s) + \mathcal{E}_{LO}\cos(\omega_{LO}(t) + \phi_{LO})|^2$

$I(t) = \mathcal{E}_0^2(t)\cos^2(\omega_s t + \phi_s) + \mathcal{E}_{LO}^2\cos(\omega_{LO}(t) + \phi_{LO}) + \mathcal{E}_0(t)\mathcal{E}_{LO}[\cos((\omega_s + \omega_{LO})t + \phi_s + \phi_{LO}) + \cos((\omega_s - \omega_{LO}) + \phi_s - \phi_{LO})]$

b. It is necessary to have $P_{LO} >> P_s$, such that $\mathcal{E}_{LO} >> \mathcal{E}_s$

Then, the photocurrent will have a frequency BW within the PD bandwidth.

$\rightarrow I(t) = \underbrace{\mathcal{E}_0^2(t)\cos^2(\omega_s t + \phi_s)}_{\text{small}} + \underbrace{\mathcal{E}_{LO}^2\cos(\omega_{LO}(t) + \phi_{LO})}_{\text{no data}} + \mathcal{E}_0(t)\mathcal{E}_{LO}[\underbrace{\cos((\omega_s + \omega_{LO})t + \phi_s + \phi_{LO})}_{\text{outside PD BW}} + \underbrace{\cos((\omega_s - \omega_{LO}) + \phi_s - \phi_{LO})}_{\text{inside PD BW}}]$

$I(t) \propto \mathcal{E}_0(t)\mathcal{E}_{LO}\cos((\omega_s - \omega_{LO})t + (\phi_s - \phi_{LO}))$

c. P_{LO} increased by 3dB $\rightarrow \mathcal{E}_{LO} = \sqrt{P_{LO}} \rightarrow I(t)$ increased by $\sqrt{3dB}$

d. Since $P_{LO} >> P_s$, $I(t)$ will increase by $\approx \sqrt{3dB}$

3. $\phi_{NL} = \gamma \cdot P_{in} \cdot L$, $\gamma = \frac{2\pi n_2}{A_{eff} \cdot \lambda}$, want $\phi_{NL} = 2\pi$

$L = \frac{2\pi}{\left(\frac{2\pi n_2}{A_{eff} \cdot \lambda}\right) \cdot P_{in}}$, $P_{in} = 6dBm = 0.004W$, $\lambda = 1550e\text{-}9m$, $A_{eff} = 5e\text{-}11m^2$, $n_2 = 2.6e\text{-}20m^2/W$, $L = \frac{2\pi}{\left(\frac{2\pi(2.6e\text{-}20m^2/W)}{5e\text{-}11m^2 \cdot 1.55e\text{-}9m}\right) \cdot 0.004W}$

$= \boxed{745.19km}$

4. QPSK $\rightarrow$ 2 bits/symbol time

a. QAM $\rightarrow$ 4 bits/symbol time $\rightarrow$ higher spectral efficiency

higher probability for bit error rate (BER) due to:

white Gaussian noise if power levels are small

nonlinear degrading effects if power levels are high

b. need more spectral efficiency $\rightarrow$ choose QAM

c. choose QPSK $\rightarrow$ reduce probability for error